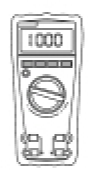
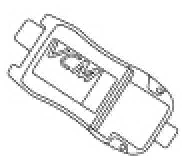
Autolamps

Ford Ecat Electronic Parts Catalog

Special Tool(s)

 ST3093-A	Fluke 77-1V Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

Exterior Lighting

The Steering Column Control Module (SCCM) monitors the headlamp switch position by sending voltage signals on multiple circuits to the headlamp switch. There is one circuit for each headlamp switch position. At any given time, one of the signal circuits is switched to ground. The SCCM sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the headlamp switch position.

If the SCCM does not detect any active inputs from the headlamp switch for 5 seconds, or if the SCCM detects multiple headlamp switch input circuits are active, the SCCM sends a message to the BCM to indicate the fault. The BCM then turns the parking lamps and headlamps on and keeps them on until the battery saver feature times out.

Refer to Exterior Lighting in the Description and Operation portion of this section for information regarding the battery saver feature.

If either situation occurs, the SCCM and the BCM **cannot** be ruled immediately as being at fault. This is normal behavior of the SCCM and the BCM design as a fault has been detected with the inputs from the headlamp switch.

The autolamp feature is enabled when the BCM receives a network message from the SCCM indicating the headlamp switch is in the AUTOLAMPS ON position.

Autolamps Feature

The BCM monitors the light sensor with a voltage signal. The light sensor input to the BCM varies with the ambient light conditions.

When the BCM receives a message from the SCCM indicating a request for the autolamps, the BCM monitors the light sensor for the ambient light condition. If the BCM determines the ambient light level is dark, the BCM supplies voltage to the exterior lamps.

If the ignition is already in RUN and the vehicle enters a dark/lighted area (such as when entering/exiting a non-lighted tunnel during the daytime), the transition from light to dark (or dark to light) needs to last 15 seconds before the BCM turns the exterior lamps on or off. This strategy prevents the exterior lamps from unnecessarily flashing on and off.

The BCM reacts quicker under extreme light conditions. If the BCM has detected a very dark condition, the exterior lamps are turned on after 1.5 seconds. If the BCM has detected a very high ambient light level, the exterior lamps are turned off after 4 seconds.

If a fault is detected with the light sensor input, the BCM turns the parking lamps and headlamps on.

Headlamps On With Wipers On Feature

When the autolamps feature is active and the front wipers are on for more than 3 seconds (except during a mist wipe or while the wipers are on to clear washer fluid during a wash condition), the exterior lamps are turned on. The exterior lamps are turned off when the ignition is changed to OFF or ACC, the headlamp switch is placed in the OFF position, or the front wipers are off for more than 30 seconds. The exception to this is when the exterior lights are on because of darkness determined by the light sensor.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Headlamp switch	• Wiring, terminals or connectors • Light sensor

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. **NOTE:** *Make sure to use the latest scan tool software release.*

   If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .
5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*

   If the scan tool does not communicate with the VCM :
   - check the VCM connection to the vehicle.
   - check the scan tool connection to the VCM .
   - refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.
6. If the scan tool does not communicate with the vehicle:
   - verify the ignition is on.
     - The air bag warning indicator prove-out confirms ignition on (other indicators may not prove ignition on). If the ignition does not turn on, refer to Section 211-05 to diagnose no power in run.
   - verify the scan tool operation with a known good vehicle.
   - refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

7. Carry out the network test.
    - If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
    - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the Body Control Module (BCM) and the Steering Column Control Module (SCCM) .
9. If the DTCs retrieved are related to the concern, refer to Diagnostic Trouble Code (DTC) Chart in this section. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.
10. If no DTCs related to the concern are retrieved, GO to Symptom Chart.

Symptom Chart

Symptom Chart

Condition	Possible Causes	Action
• The autolamps are inoperative	• Light sensor • Body Control Module (BCM)	• GO to Pinpoint Test G.
• The autolamps are on continuously	• Wiring, terminals or connectors • Light sensor • Headlamp switch • Steering Column Control Module (SCCM) • BCM	• GO to Pinpoint Test H.

Pinpoint Tests

Pinpoint Test G: The Autolamps Are Inoperative

Refer to Wiring Diagrams Cell 85 , Headlamps/Autolamps for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) sends voltage signals to the headlamp switch through the headlamp switch input circuits (off, parking lamps, headlamps, autolamps). At any given time, the headlamp switch routes one of the input circuits to ground, indicating the headlamp switch position to the SCCM .

The SCCM sends the status of the headlamp switch position to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) .

The autolamps feature is enabled when the BCM receives a message from the SCCM indicating the headlamp switch is in the AUTOLAMPS ON position.

When the BCM enables the autolamps feature, the BCM monitors the light sensor input. The light sensor determines the amount of light based on the input received from the ambient lighting conditions. When the ambient level has reached a point (determined by the internal programming of the BCM), the BCM provides voltage to the exterior lamps.

Any fault detected from the light sensor or headlamp switch inputs causes the exterior lamps to be turned on.

This pinpoint test is intended to diagnose the following:

- Light sensor

- BCM

PINPOINT TEST G : THE AUTOLAMPS ARE INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

G1 CHECK THE MANUAL HEADLAMP OPERATION

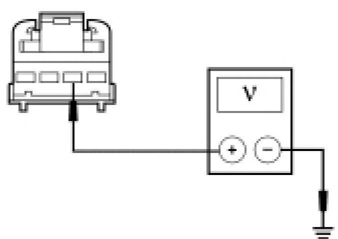
- Place the headlamp switch in the HEADLAMPS ON and OFF positions.

Do the headlamps operate correctly?

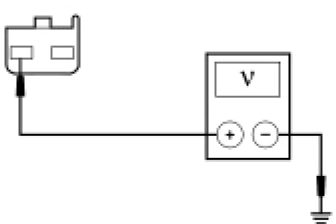
Yes	GO to G2.
No	REFER to Headlamps in this section.

G2 CHECK FOR VOLTAGE TO THE LIGHT SENSOR

- Ignition OFF.
- Disconnect: Light Sensor C287 (Manual Temperature Control) or C286 (Automatic Temperature Control) .
- Ignition ON.
- Place the headlamp switch in the AUTOLAMPS ON position.
- For vehicles with automatic temperature control, measure the voltage between the light sensor C286 Pin 2, circuit VLF14 (BU), harness side and ground.



- N0057372
- For vehicles with manual temperature control, measure the voltage between the light sensor C287 Pin A, circuit VLF14 (BU/BN), harness side and ground.



- N0057373

Is the voltage approximately 5 volts?

Yes	INSTALL a new light sensor. REFER to Light Sensor in this section. TEST the system for normal operation.

No	GO to G3.

G3 CHECK FOR CORRECT BCM OPERATION

• Disconnect all the BCM connectors.
• Check for:
  ▪ corrosion
  ▪ damaged pins
  ▪ pushed-out pins
• Connect all the BCM connectors and make sure they seat correctly.
• Operate the system and verify the concern is still present.
 	
Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test H: The Autolamps Are On Continuously

Refer to Wiring Diagrams Cell 85 , Headlamps/Autolamps for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) sends voltage signals to the headlamp switch through the headlamp switch input circuits (off, parking lamps, headlamps, autolamps). At any given time, the headlamp switch routes one of the input circuits to ground, indicating the headlamp switch position to the SCCM .

The SCCM sends the status of the headlamp switch position to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) .

The autolamp feature is enabled when the BCM receives a message from the SCCM indicating the headlamp switch in the AUTOLAMPS ON position.

When the BCM enables the autolamps feature, the BCM monitors the light sensor input. The light sensor determines the amount of light based on the input received from the ambient lighting conditions. When the ambient level has reached a point (determined by the internal programming of the BCM), the BCM provides voltage to the exterior lamps.

If the SCCM detects multiple or no active headlamp switch inputs, it sends a message to the BCM to indicate a fault with the headlamp switch input.

The BCM defaults the exterior lamps on when the BCM detects any of the following conditions:

- a fault with the light sensor input
- a message from the SCCM indicating a fault with the headlamp switch input
- the BCM loses communication with the SCCM

Autolamp System DTCs

- DTC B1A85:11 (Ambient Light Sensor: Circuit Short To Ground) — This DTC sets when the BCM detects a short to ground from the light sensor input circuit.
- DTC B1A85:13 (Ambient Light Sensor: Circuit Open) — This DTC sets when the BCM detects an open or short to voltage from the light sensor input circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Light sensor
- Headlamp switch
- SCCM
- BCM

PINPOINT TEST H : THE AUTOLAMPS ARE ON CONTINUOUSLY

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Make sure the HVAC control operates correctly before diagnosing the autolamp system.*

H1 CHECK THE SCCM AUTOLAMP ON REQUEST PID

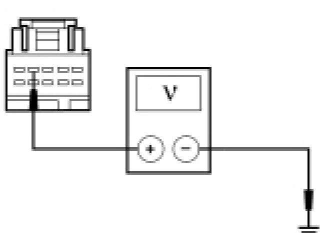
NOTE: *Make sure the headlamp switch is lined up in the correct position when monitoring the PID.*

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: SCCM DataLogger .
- While placing the headlamp switch in the AUTOLAMPS ON and then OFF position, monitor the SCCM headlamp switch PID (EXT_LMP_SW).

Does the PID agree with the headlamp switch when it is in the AUTOLAMPS ON position and indicate it is not in the AUTOLAMPS ON position when the headlamp switch is in the OFF position?

Yes	GO to H5.
No	GO to H2.

H2 CHECK FOR A VOLTAGE SIGNAL TO THE HEADLAMP SWITCH

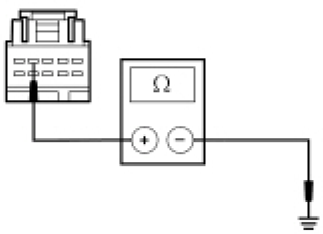
- Ignition OFF.
- Disconnect: Headlamp Switch C205.
- Measure the voltage between the headlamp switch C205 Pin 4, circuit CLF19 (VT), harness side and ground.
- 

N0099677

Is the voltage greater than 5 volts?

Yes	INSTALL a new headlamp switch. REFER to Headlamp Switch in this section. TEST the system for normal operation.
No	GO to H3.

H3 CHECK AUTOLAMPS REQUEST INPUT CIRCUIT FOR A SHORT TO GROUND

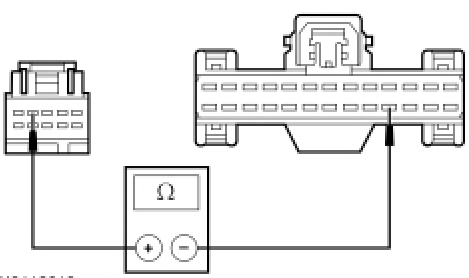
- Disconnect: SCCM C2414A .
- Measure the resistance between the headlamp switch C205 Pin 4, circuit CLF19 (VT), harness side and ground.
- 

N0083964

Is the resistance greater than 10,000 ohms?

Yes	GO to H4.
No	REPAIR circuit CLF19 (VT) for a short to ground. TEST the system for normal operation.

H4 CHECK THE AUTOLAMPS REQUEST INPUT CIRCUIT FOR AN OPEN

- Measure the resistance between the headlamp switch C205 Pin 4, circuit CLF19 (VT), harness side and the SCCM C2414A Pin 17, circuit CLF19 (VT), harness side.
- 

N0112016

Is the resistance less than 5 ohms?

Yes	GO to H14.
No	REPAIR circuit CLF19 (VT) for an open. TEST the system for normal operation.

H5 CHECK THE SCCM HEADLAMP SWITCH PIDS

NOTE: *Make sure the headlamp switch is aligned in the correct position when monitoring the PID.*

- Enter the following diagnostic mode on the scan tool: SCCM DataLogger .

- While moving the headlamp switch through all positions (OFF, PARKING LAMPS and HEADLAMPS), monitor the SCCM headlamp switch input PID (EXT_LMP_SW).

Do the headlamp switch positions agree with the PID?

Yes	GO to H6.
No	GO to Pinpoint Test C.

H6 USE THE RECORDED DTCS FROM THE BCM SELF-TEST

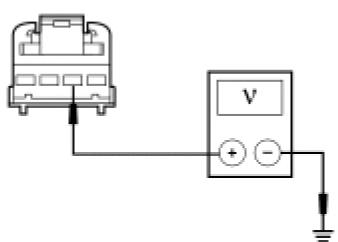
- Check the recorded results from the BCM self-test.

Is DTC B1A85:13 or DTC B1A85:11 present?

Yes	For DTC B1A85:13, GO to H8. For DTC B1A85:11, GO to H12.
No	GO to H7.

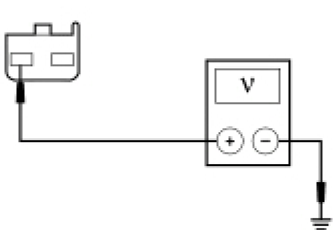
H7 CHECK FOR VOLTAGE TO THE LIGHT SENSOR (NO DTC)

- Ignition OFF.
- Disconnect: Light Sensor C287 (Manual Temperature Control) or C286 (Automatic Temperature Control) .
- Ignition ON.
- Place the headlamp switch in the AUTOLAMPS ON position.
- On vehicles with automatic temperature control, measure the voltage between the light sensor C286 Pin 2, circuit VLF14 (BU), harness side and ground.



- N0057372

- On vehicles with manual temperature control, measure the voltage between the light sensor C287 Pin A, circuit VLF14 (BU/BN), harness side and ground.

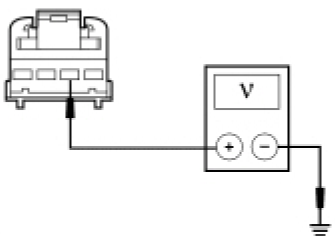


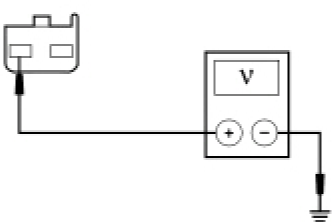
- N0057373

Is the voltage approximately 5 volts?

Yes	INSTALL a new light sensor. REFER to Light Sensor in this section. TEST the system for normal operation.
No	GO to H15.

H8 CHECK FOR VOLTAGE TO THE LIGHT SENSOR (DTC B1A85:13)

- Ignition OFF.
- Disconnect: Light Sensor C287 (Manual Temperature Control) or C286 (Automatic Temperature Control) .
- Ignition ON.
- Place the headlamp switch in the AUTOLAMPS ON position.
- On vehicles with automatic temperature control, measure the voltage between the light sensor C286 Pin 2, circuit VLF14 (BU), harness side and ground.
- 

N0057372
- On vehicles with manual temperature control, measure the voltage between the light sensor C287 Pin A, circuit VLF14 (BU/BN), harness side and ground.
- 

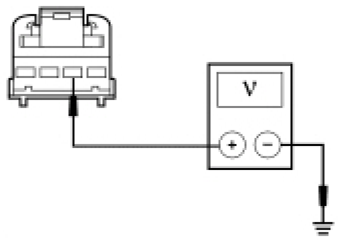
N0057373

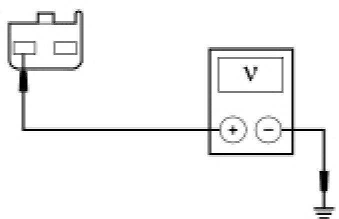
Is the voltage approximately 5 volts?

Yes	GO to H11.
No	GO to H9.

H9 CHECK THE LIGHT SENSOR INPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM C2280B .

- Ignition ON.
- On vehicles with automatic temperature control, measure the voltage between the light sensor C286 Pin 2, circuit VLF14 (BU), harness side and ground.
- 

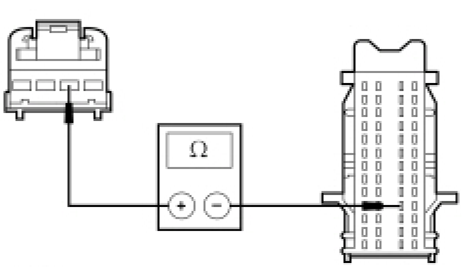
  N0057372
- On vehicles with manual temperature control, measure the voltage between the light sensor C287 Pin A, circuit VLF14 (BU/BN), harness side and ground.
- 

  N0057373

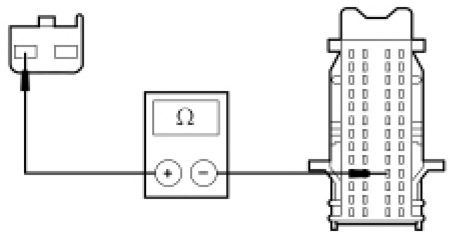
Is any voltage present?

Yes	REPAIR circuit VLF14 (BU/BN) or VLF14 (BU) for a short to voltage. TEST the system for normal operation.
No	GO to H10.

H10 CHECK THE LIGHT SENSOR INPUT CIRCUIT FOR AN OPEN

- Ignition OFF.
- On vehicles with automatic temperature control, measure the resistance between the light sensor C286 Pin 2, circuit VLF14 (BU), harness side and the BCM C2280B Pin 30, circuit VLF14 (BU), harness side.
- 

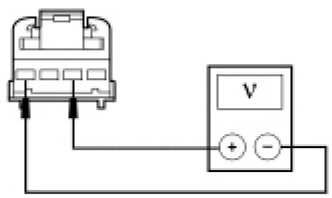
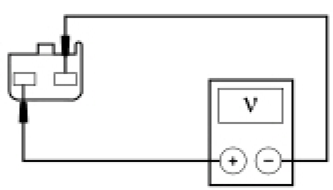
  N0112017
- On vehicles with manual temperature control, measure the resistance between the light sensor C287 Pin A, circuit VLF14 (BU/BN), harness side and the BCM C2280B Pin 30, circuit VLF14 (BU/BN), harness side.

- 
  N0112018

Is the resistance less than 5 ohms?

Yes	GO to H15.
No	REPAIR circuit VLF14 (BU/BN) or VLF14 (BU) for an open. TEST the system for normal operation.

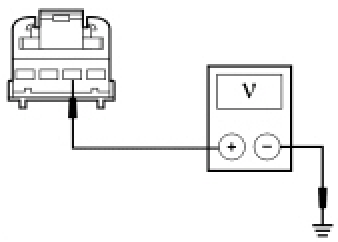
H11 CHECK THE LIGHT SENSOR GROUND CIRCUIT FOR AN OPEN

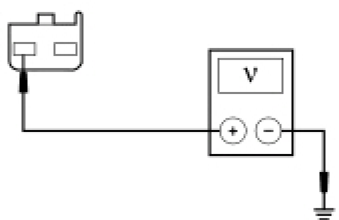
- For vehicles with automatic temperature control, measure the voltage between the light sensor C286 Pin 2, circuit VLF14 (BU), harness side and the light sensor C286 Pin 4, circuit RH111 (GY/BU), harness side.
- 
  N0088170
- For vehicles with manual temperature control, measure the voltage between the light sensor C287 Pin A, circuit VLF14 (BU/BN), harness side and the light sensor C287 Pin B, circuit RH111 (GY/BU), harness side.
- 
  N0088171

Is the voltage approximately 5 volts?

Yes	INSTALL a new light sensor. REFER to Light Sensor in this section. TEST the system for normal operation.
No	REPAIR circuit RH111 (GY/BU) for an open. TEST the system for normal operation.

H12 CHECK FOR VOLTAGE TO THE LIGHT SENSOR (DTC B1A85:11)

- Ignition OFF.
- Disconnect: Light Sensor C287 (Manual Temperature Control) or C286 (Automatic Temperature Control) .
- Ignition ON.
- Place the headlamp switch in the AUTOLAMPS ON position.
- On vehicles with automatic temperature control, measure the voltage between the light sensor C286 Pin 2, circuit VLF14 (BU), harness side and ground.
- 

N0057372
- On vehicles with manual temperature control, measure the voltage between the light sensor C287 Pin A, circuit VLF14 (BU/BN), harness side and ground.
- 

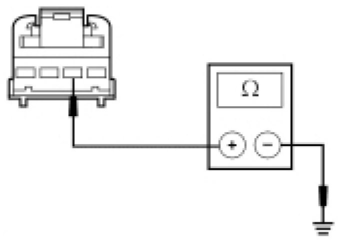
N0057373

Is the voltage approximately 5 volts?

Yes	INSTALL a new light sensor. REFER to Light Sensor in this section. TEST the system for normal operation.
No	GO to H13.

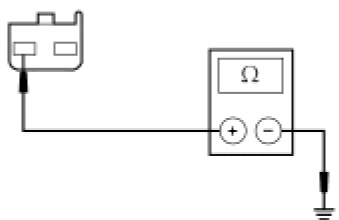
H13 CHECK THE LIGHT SENSOR INPUT CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: BCM C2280B .
- On vehicles with automatic temperature control, measure the resistance between the light sensor C286 Pin 2, circuit VLF14 (BU), harness side and ground.



- N0057374

- On vehicles with manual temperature control, measure the resistance between the light sensor C287 Pin A, circuit VLF14 (BU/BN), harness side and ground.



- N0057375

Is the resistance greater than 10,000 ohms?

Yes	GO to H15.
No	REPAIR circuit VLF14 (BU/BN) or VLF14 (BU) for a short to ground. TEST the system for normal operation.

H14 CHECK FOR CORRECT SCCM OPERATION

- Disconnect all the SCCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the SCCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new SCCM . REFER to Section 211-05. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

H15 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:

- corrosion
- damaged pins
- pushed-out pins

- Connect all the BCM connectors and make sure they seat correctly.

- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.